1. The Degradation Problem:

when we increase a network depth, the training error increases even if it seems to have healthy gradient descent values in this model.
if it is only the increase of test error, then that would be an indication of the Overfitting problem.
however, the increasing training error indicates that this deeper model simply doesn't manage to learn what it should be able to learn, and it doesn't result from the Overfitting problem.

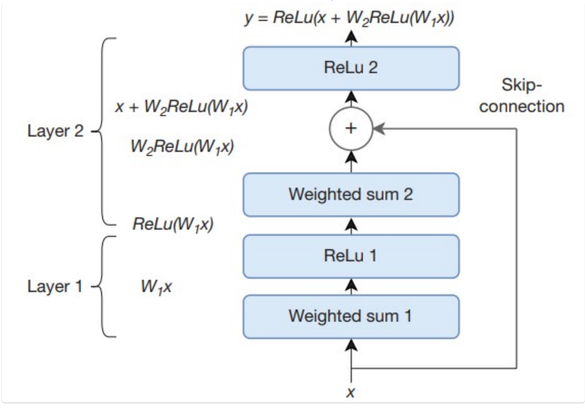
2. ResNet:

A. a ResNet solves the Degradation problem by using a mechanism known as the Skip Connection.
that make it easier for a neuron network to learn more close to the identity function (== f(x)=x).

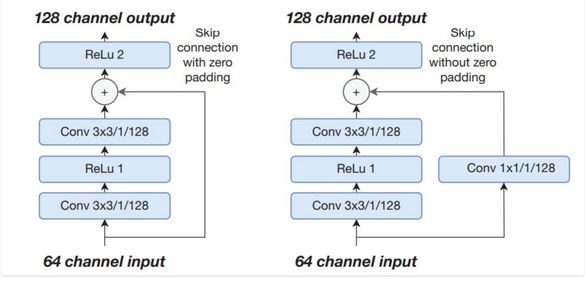
B. the skip connection:

⇒ it contains two stacked convolutional layers with an additional skip connection that bypasses these two convolutional layers.
the input data to the first convolutional layer is added to the weighted sum function of the second convolutional layer,
and then both of them are fed through the activation function of the second convolutional layer.

⇒ x = the input data to the first convolutional layer.
Weighted Sum 1 = the weighted sum function of the first convolutional layer $= W_1 \cdot x$.
ReLU 1 = the activation function of the first convolutional layer $= ReLU(W_1 \cdot x)$.
Weighted Sum 2 = the weighted sum function of the second convolutional layer $= W_2 \cdot ReLU(W_1 \cdot x)$.
ReLU 2 = the activation function of the second convolutional layer $= ReLU(x + W_2 \cdot ReLU(W_1 \cdot x))$.



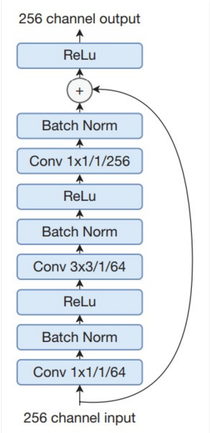
⇒ if a skip connection for convolutional layers encounters with the more num of output channels in the last convolutional layer than the num of input channels in the first convolutional layer,
the case is represented as the dashed skip connection, and there are two ways to implement a dashed skip connection.
the left implementation in the figure below:
we can execute a skip connection to only half of the output channels in the last convolutional layer (==128/2==64) to correspond the num of input channels in the first convolutional layer (=64),
and then using zero to pads the surplus output channels in the last convolutional layer (==128-64=64) without applying a skip connection.
the right implementation in the figure below:
we can also expand the num of input channels in the first convolutional layer (=64) by using 1×1/1/XX convolutional layer (=1×1/1/128) to match the num of output channels in the last convolutional layer (=128).



C. a typical residual module in a ResNet can be divided into two types:

⇒ the basic module: it is used in a shallower ResNet (Ex: ResNet-18, ResNet-34), and it is composed of two stacked 3×3/1/XX convolutional layers.

⇒ the bottleneck module: it is used in a deeper ResNet (Ex: ResNet-50, ResNet-152), and it is composed of three stacked 1×1/1=2/XX → 3×3/1/XX → 1×1/1/XX convolutional layers.
we can use the 1×1/1=2/XX convolutional layer to temporarily reduce the num of input channels in the 3×3/1/XX convolutional layer so as to reduce the num of required weights in the 3×3/1/XX convolutional layer,
and then we finally use another 1×1/1/XX convolutional layer to increase the num of over channels in a bottleneck module again.



D. a bottleneck module periodically introduces the 1×1/2/XX convolutional layer to halve the width and height dimensions of input data for the next convolutional layer.

E. a bottleneck module also periodically introduces the 1×1/1=2/XX convolutional layer with the num of output channels (= num of distinct kernels = XX) doubling its num of input channels
so as to double the input channel dimension of input data for the next convolutional layer.

F. each neuron of all convolutional layers in a ResNet use ReLU as an activation function.

H.

Table 8-3 ResNet Architecture*

34 LAYER	50 LAYER	152 LAYER
Conv 7×7/2/64 Max pool 3×3/2		
Conv [3×3/1/64, 3×3/1/64] ×3	Conv [1×1/1/64, 3×3/1/64, 1×1/1/256] ×3	Conv [1×1/1/64, 3×3/1/64, 1×1/1/256] ×3
Conv [3×3/2/128, 3×3/1/128] ×1	Conv [1×1/2/128, 3×3/1/128, 1×1/1/512] ×1	Conv [1×1/2/128, 3×3/1/128, 1×1/1/512] ×1
Conv [3×3/1/128, 3×3/1/128] ×3	Conv [1×1/1/128, 3×3/1/128, 1×1/1/512] ×3	Conv [1×1/1/128, 3×3/1/128, 1×1/1/512] ×7
Conv [3×3/2/256, 3×3/1/256] ×1	Conv [1×1/2/256, 3×3/1/256, 1×1/1/1,024] ×1	Conv [1×1/2/256, 3×3/1/256, 1×1/1/1,024] ×1
Conv [3×3/1/256, 3×3/1/256] ×5	Conv [1×1/1/256, 3×3/1/256, 1×1/1/1,024] ×5	Conv [1×1/1/256, 3×3/1/256, 1×1/1/1,024] ×35
Conv [3×3/2/512, 3×3/1/512] ×1	Conv [1×1/2/512, 3×3/1/512, 1×1/1/2,048] ×1	Conv [1×1/2/512, 3×3/1/512, 1×1/1/2,048] ×1
Conv [3×3/1/512, 3×3/1/512] ×2	Conv [1×1/1/512, 3×3/1/512, 1×1/1/2,048] ×2	Conv [1×1/1/512, 3×3/1/512, 1×1/1/2,048] ×2
Avg pool 7×7/1		
FC softmax 1000		